

$$\begin{aligned}x_1 - 2x_2 + x_3 &= 0 \\x_2 - 4x_3 &= 4 \\30x_3 &= -30\end{aligned}$$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 30 & -30 \end{bmatrix}$$

Now, multiply equation 3 by $\frac{1}{30}$ in order to obtain 1 as the coefficient for x_3 .
(This calculation will simplify the arithmetic in the next step).

$$\begin{aligned}x_1 - 2x_2 + x_3 &= 0 \\x_2 - 4x_3 &= 4 \\x_3 &= -1\end{aligned}$$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

The new system has a Triangular form

$$\begin{aligned}x_1 - 2x_2 + x_3 &= 0 \\x_2 - 4x_3 &= 4 \\x_3 &= -1\end{aligned}$$

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

Eventually, you want to eliminate the $-2x_2$ term from equation 1, but it is more efficient to use the x_3 in equation 3 first, to eliminate the $-4x_3$ and $+x_3$ terms in equations 2 and 1.

The two "mental" calculations are:

$$\begin{array}{l} 4 \cdot [\text{equation 3}] \\ + [\text{equation 2}] \\ \hline [\text{new equation 2}] \end{array} \quad \begin{array}{l} 4x_3 = -4 \\ x_2 - 4x_3 = 4 \\ \hline x_2 = 0 \end{array}$$

$$\begin{array}{l} -1 \cdot [\text{equation 3}] \\ + [\text{equation 1}] \\ \hline [\text{new equation 1}] \end{array} \quad \begin{array}{l} -x_3 = 1 \\ x_1 - 2x_2 + x_3 = 0 \\ \hline x_1 - 2x_2 = 1 \end{array}$$

It is convenient to combine the results of these two operations:

$$\begin{aligned}x_1 - 2x_2 &= 1 \\x_2 &= 0 \\x_3 &= -1\end{aligned} \quad \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

Now, having cleaned out the column above the x_3 in equation 3, move back to the x_2 in equation 2 and use it to eliminate the $-2x_2$ above it. Because of the previous work with x_3 , there is now no arithmetic involving x_3 terms.

Add 2 times equation 2 to equation 1 and obtain the system:

$$\begin{aligned}x_1 &= 1 \\x_2 &= 0 \\x_3 &= -1\end{aligned} \quad \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$